

Kuricheti Prerana

preranakuricheti18@gmail.com | +91 89193 42535 | [LinkedIn](#) | [GitHub](#)

SUMMARY

AI Engineering student (CGPA 9.06/10) with hands-on experience building production-grade Retrieval-Augmented Generation (RAG) pipelines, multi-agent systems, and LLM-powered applications. Skilled in hybrid retrieval, vector databases, and backend API development, with a track record of peer-reviewed publications and end-to-end project ownership. Quick to adopt emerging AI agent frameworks and thrives in fast-paced, product-focused environments.

EXPERIENCE

AI Engineer Intern, RBG.ai

April 2026 – June 2026

- Built a production-grade multilingual RAG system enabling cross-language (English–Spanish) document question-answering, integrating LLM inference (LLaMA via Groq), dynamic prompting, and source attribution.
- Designed a hybrid retrieval pipeline combining BM25, vector search, and cross-encoder re-ranking, and optimized performance using query expansion, caching, and efficient chunking techniques.
- Developed a multi-stage PDF ingestion pipeline supporting OCR, table extraction, and image understanding for both scanned and native documents.
- Implemented a multi-tenant architecture using Qdrant and built FastAPI-based REST services for document ingestion, querying, and intermediate inspection.

PROJECTS

VioletAI – Stock Research Copilot

Feb 2026 – Mar 2026

- Built a multi-agent AI system for automated equity research using LLMs, RAG, sentiment analysis, and forecasting to generate explainable insights with real-time dashboards.

AI Startup Idea Validation Agent

Feb 2026 – Mar 2026

- Developed an AI agent that evaluates startup ideas by performing automated market research, competitor analysis, and LLM-based feasibility scoring.

Hinglish Sentiment Analysis for Indian Election Tweets

Dec 2025 – Mar 2026

- Built a Hinglish-language sentiment and text classification model using NLP and MuRIL-based transformer architectures.

Orion – LLM-Powered Intelligent Assistant

Jul 2025 – Dec 2025

- Developed an LLM-based RAG system for accurate, document-grounded question-answering using chunking, embeddings, semantic search, and OCR for PDFs.

RSMA-Based UAN Channel Modelling for Coral Reef Monitoring via ML

Mar 2025 – May 2025

- Developed an autoencoder-based embedding pipeline combining deep learning and Random Forest to generate confidence-weighted rankings, evaluated using analytical and Monte Carlo simulations.

EDUCATION

B.Tech in Artificial Intelligence, Amrita Vishwa Vidyapeetham, Coimbatore

2023 – 2027

CGPA: 9.06 / 10

TECHNICAL SKILLS

Programming Languages: Python, C++, C, JavaScript, Scala

AI / ML: Machine Learning, Deep Learning, Computer Vision, Signal & Image Processing

Core CS & Software Engineering: Object-Oriented Design, DSA, REST API Design & Integration, Debugging & Performance Optimization, Code Review, Web Development(HTML/CSS)

LLM / GenAI: RAG Pipelines, LLMs, Prompt Engineering, Vector Databases, LangChain, CrewAI, Multi-Agent Systems

Backend & Systems: FastAPI, REST APIs, Qdrant, ChromaDB, PostgreSQL, MongoDB

Tools & Infrastructure: Docker, Git & GitHub, Cloud Computing

AI Tools: ChatGPT, Claude, GitHub Copilot

PUBLICATIONS

- Kuricheti Prerana, et al., “Automated Crop Health Monitoring using NDVI and Machine Learning Techniques,” International Society for Photogrammetry and Remote Sensing (ISPRS) Conference, 2025.
- Kuricheti Prerana, et al., “Investigation of EEGNet and its Variants for ADHD Detection,” ICBSII, Chennai, 2025.